

ESTELA KATCHEN

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Education

University of Western Ontario & Ivey Business School

London, ON | 2024-present

Dual degree: BESC (Mechatronic Systems Engineering) & HBA (Business Administration)

• Dean's Honour List, 2025-2026

Experience

Summer AI Research Assistant

May 2026 – Present

Yang's AiLearn Lab, Prof. Yimin Yang, Western University — London, ON

- Contributing to AI/machine learning (ML) research for mammogram analysis, supporting development of a domain-specific vision-language model to improve clinical interpretability and generalizability.
- Supporting data preprocessing, model training, and performance evaluation in Python and PyTorch across cancer classification and detection tasks on medical imaging datasets. Improved classification accuracy by 6% compared to baseline.

Product Analyst Intern

May – Aug 2025

Wahi – RPS (Brookfield Portfolio Company) — Toronto, ON

- Led competitive benchmarking across 6+ valuation technology platforms and 7 product dimensions, identifying gaps in user experience (UX), workflows, onboarding, support, and client reporting.
- Interviewed 20+ clients and stakeholders, including appraisers and app users, and analyzed Datadog usage data to identify workflow friction points and optimize user experience.
- Produced executive-ready strategy materials, including feature matrices and UX teardown reports; delivered 10+ prioritized recommendations adopted by the product team to improve workflow efficiency and client experience.

Leadership & Involvement

Sponsorship Ambassador (2026-27) | VP Mentorship (2025-26) | First-Year Rep (2024-25)

Sept 2024 – Present

Western Women in Engineering (WiE)

- Securing corporate sponsorships for programming across a 2,000+ member organization; managing partner relationships and negotiating event deliverables.
- Led mentorship programs supporting 600+ students; designed matching frameworks and managed communications, logistics, and participant feedback.

VP Competitions — Western Engineering Competition (WEC)

May 2026 – Present

- Directing end-to-end planning for WEC 2026: scoping competition formats, designing judging rubrics, and coordinating cross-team logistics across 5+ categories.

Data Team Developer — 3DWestern

May 2026 – Present

- Building data preprocessing pipelines and benchmarking open-source vision-language models to support ML-based 3D print quality classification.

Engineering Orientation Week Leader (Soph)

Aug 2025 – Mar 2026

- Selected to mentor incoming engineering students through Orientation Week programming, supporting first-year transition, community-building, and student engagement.

Projects

Colour-Sorting Robot

Mar – Apr 2026

- Designed and built an ESP32/C++ robot to autonomously detect and sort coloured objects, coordinating control logic across movement, sensing, and sorting subsystems.

Autonomous Retrieval Robot

Feb 2026

- Built an ESP32/C++ mobile robot with autonomous navigation, sensor-driven object detection, mechanical claw actuation, and Git-based version control.

Additional Experience

Sailing Instructor & Race Team

June 2023 – Aug 2024

Royal Canadian Yacht Club / OHSF – Instructed youth sailing and raced 29ers nationally.

Swim Instructor & Lifeguard

Jan 2022 – Aug 2024

Miles Nadal JCC – Delivered swim instruction and maintained safety for up to 30 swimmers per shift.

Violinist & Volunteer Instructor

2012 – 2024

Royal Conservatory of Music / The Hammer Band – RCM Grade 9; performed at Koerner Hall and mentored junior musicians.

Skills

Technical: Java, C/C++, Python, PyTorch, MATLAB, Git, SolidWorks (CSWA), Onshape, Excel, Datadog

Languages: Spanish (fluent), French (intermediate)